KOÇ UNIVERSITY

College of Engineering
COMP 491 – Computer Engineering Design
Project Proposal

Phoenix
Simulation-Driven Development of an Autonomous Search-and-Rescue Navigation System

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Participant Information:

Name	ID	Email	Phone
Ömer Maraş	82096	ojeblawı22@ku.edu.tr	+90 537 857 65 99
Mohamad Alomar	82215	malomar22@ku.edu.tr	+90 505 010 09 00
Abdul Rahman Khaled Abu Audeh	82348	aaudeh22@ku.edu.tr	+90 535 933 76 97

Project Advisor: Fatma Güney

Abstract

Search and Rescue Vehicles are an ever-growing part in lowering the fatality rates of missing individuals especially in disaster zones. Current autonomous driving works and studies have focused on subsets of the full driving task however, implementations of such autonomous systems in unstructured dynamically evolving environments remain limited, largely due to challenges in reliable perception, accurate localization, and real-time decision making under uncertainty. Our project aims to take a step in this direction by developing a simulation led work pipeline that bridges theoretical autonomous driving research with deployable real-time search and rescue applications.

This project proposes the design and implementation of a simulation-driven Autonomous Search and Rescue Vehicle (SRV) able to navigate a maze, locate a Wi-Fi emitting target, and construct a path from start position to target position without human intervention. The system leverages LiDAR based Simultaneous Localization and Mapping (SLAM), camera-based object detection, and IEEE 802.11mc Fine Timing Measurement to determine the location of a target. It utilizes statistical preprocessing, and a particle filter framework to mitigate reflection and measurement noise. The system performs global path planning using a heuristic algorithm over a SLAM-generated occupancy grid while avoiding dynamic obstacles detected in real-time. The design is based on a 4-stage prototyping approach: (1) Evaluation of basic FTM ranging, (2) implementation of full-fledged SLAM, (3) computer vision model training and testing, and (4) dynamic path planning and obstacle avoidance for autonomous missions. All stages will be preceded by simulation-based validation to ensure performance robustness prior to physical deployment. Experiments will be conducted across different maze-based environments to analyse localization accuracy, convergence behaviour, and environmental sensitivity. The outcome of this project will be a fully autonomous SRV, a quantitative assessment of Wi-Fi FTM feasibility for indoor SAR applications, and a dynamically responsive navigation framework capable of real-time obstacle detection, multi-sensor fusion, and optimal path planning under uncertain and cluttered conditions.

We hope that our autonomous SRV will serve as a modular and extensible research testing ground for disaster-response scenarios and hope to provide a reproducible framework for evaluating multi-sensor fusion, RF-based localization, and autonomous navigation strategies in GPS-denied and dynamically evolving environments.

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Section 1 Introduction

1.1 Concept

Search and Rescue (SAR) operations in indoor and GPS denied environments is a significant challenge and an important area of development. In a structured maze-like environment, such as industrial facilities, underground tunnels, or collapsed buildings, autonomous systems can excel since long range manual control is unreliable. Traditional Search and Rescue Vehicles (SRV) rely on exploration-based mapping, which is inefficient. This project proposes a hybrid approach that combines autonomous mapping and Wi-Fi based homing to reduce search time and improve performance.

The core is the development of a SRV capable of navigating structured mazes, constructing a map using LiDAR based SLAM, localizing a Wi-Fi carrying target using Fine Time measurement, identifying the target through computer vision–based image processing, and planning optimal paths for autonomous mission execution. Round Trip Time measurements obtained with FTM provide distance estimates between SRV and beacon. Unfortunately, indoor RTT is susceptible to multi-path distortion, bias, and interference. To address these issues the system will use statistical methods and particle filter-based frameworks to localize the target. The particle filter takes RTT measurements from the moving SRV and creates a probability distribution of possible locations. The reason behind choosing Wi-Fi is that almost all mobile devices nowadays support this FTM feature [9].

The development of the SRV will be modular, and simulation driven. Each component of the system will first be validated in a controlled simulation environment to evaluate performance and robustness and will then be deployed and tested on the physical platform to verify real-world reliability and environmental compatibility. The final system will be able to navigate autonomously, home towards a target, and identify it using a camera. Then construct a map to the maze exit and potentially guide the target or call on a rescue team. Beyond these deliverables, the project aims to analyse experimentally the feasibility of FTM for indoor localization. Controlled trials across multiple mazes with different levels of interference will be run; This will provide quantitative data that can be analysed for metrics such as localization error, convergence stability, and environmental stability of FTM RTT.

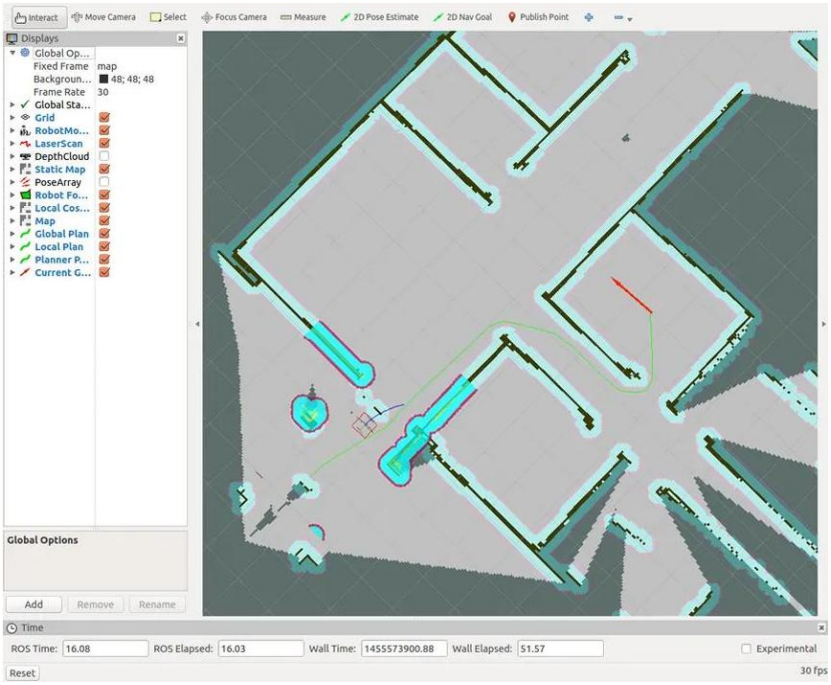


Figure 1: SLAM Map from ROS Interface [8]

1.2 Objectives

Upon completion of this project, we will have a fully autonomous Search and Rescue Vehicle that is able to navigate through a maze, locate and identify a target, then build a route from start to target. We also aim to investigate the viability of using Wi-Fi Fine Timing Measurement (FTM) combined with statistical analysis and signal processing frameworks to locate a Wi-Fi carrying target within an interference free maze system. We would also like to investigate the effects of distance, burst packet size, and reflectors on the performance for position estimation of the target.

The first step in our project will be obtaining all necessary equipment and materials. Some of the main components include: ESP32 S3, RPLIDAR C1 LiDAR, Raspberry Pi, Camera, motor drivers, and motors. While waiting for the components to arrive, we will be designing the system on simulators such as ROS2, Gazebo, and CARLA to collect data on the SRVs operation and test basic movement. We will also be sourcing either a moulded or 3D printed chassis for the system to be built in. We will build our first prototype with the aim to test the manual control and manoeuvring of our vehicle before going into autonomous missions.

Upon arrival of the main sensing and actuation hardware, we will create our prototype with ultrasonic sensors for obstacle avoidance, this will allow us to do mock tests in maze environments. Additionally, testing of FTM viability can begin and can be used to home the SRV onto the target while our LiDAR sensor arrives.

Once the LiDAR arrives, the hypersonic sensors will be replaced, and a second prototype will be built. This prototype will leverage the LiDAR, wheel encoders and potentially an IMU to provide Simultaneous Localization and Mapping (SLAM) capabilities to the vehicle. SLAM is needed to allow the SRV to detect and map all obstacles surrounding it, including walls; This map will allow us to build a route from the start of the maze to the target. We will also conduct simulations in ROS and MATLAB to test the functionality of our algorithms before this prototype's deployment. Once the target is found, a camera will utilize image processing to determine if it is indeed the desired target or not and if it is the optimum path to it from the starting point will be calculated; The target will be informed of what the SRV came to do using a speaker.

To test the viability of FTM as well as test what effects it, we plan to build 3 mazes and run 5-10 tests per maze. All mazes will be built out of the same material, cardboard or wood, and will have the same layout. The first maze will be a base maze used to test some of the basic metrics. The second maze will specifically be built to test for performance in a more complex and realistic maze environment; This will be done by adding metal/reflective surfaces and surfaces able to block 2.4GHz signals. The last maze will be made to test the interference of external Wi-Fi sources on the device's ability to detect the target, or, in other terms, the effect of having the 2.4GHz channel full on our ability to localize the target. We need to research how this can be done legally without the use of Wi-Fi jammers.

We aim to have our answer to FTM viability and a working prototype I by MDR. Then, by FDR have a SLAM capable SRV as well as answers to most of the stats described above. Additionally, if time and budget permit, we would like to have the robot transmit the mapped route to the 'rescue crew' awaiting on standby or alternatively guide the user to safety.

1.3 Background

Autonomous search-and-rescue (SAR) robotics is an increasingly important area of research due to the limitations of human responders in high-risk hazardous environments such as collapsed buildings, underground tunnels, disaster-affected infrastructure [3]. These environments are, often, GPS-denied, visually inaccessible, and structurally ambiguous, making manual navigation risky and unreliable. As a result, autonomous systems have risen amongst proposed solutions as a way to reduce search time, improve situational awareness, and lower responder risk. Modern automated driving navigation systems aim to address these issues through perception, mapping, localization, and planning pipelines that allow vehicles to sense its surroundings and navigate without human intervention [1].

A key motivation for autonomy is safety; human errors account for a vast majority of driving errors with

studies estimating approximately 94% of driving accidents being caused by human error [1]. Even though the statistics are from road transportation, this implication extends to emergency robotics. Autonomous systems and robots can operate continuously in stressful and hazardous environments where human perception limitations degrade performance. As a result, autonomous navigation technologies have been widely explored as tools to assist disaster response, alleviation, and rescue operations.

Autonomous Navigation Architecture

State-of-the-art autonomous robots follow a modular architecture composed of several core subsystems: perception, localization and mapping, planning/decision making, and control [1]. Sensor data is first processed to understand the environment, after which the robot estimates its position, predicts navigable space, and finally generates actuator commands. This structure is widely adopted because it breaks down a highly complex problem into manageable sub-problems and allows individual components to be independently developed and validated [1]. Recent research emphasizes that these modules must support the main goal of autonomy which includes planning safe motion. Modern frameworks integrate perception, prediction, and mapping so that all extracted information contributes directly to navigation decisions rather than being treated as isolated tasks [4]. This concept is particularly relevant in search-and-rescue environments, where the robot must not only detect obstacles but also infer safe routes toward a target whose position is uncertain.

Mapping and Localization Challenges

For autonomous navigation to function, the system must know both where it is and what the environment looks like. Simultaneous Localization and Mapping (SLAM) techniques solve this problem by constructing a map while estimating the robot's pose. Mapping is especially critical because, unlike humans, robots depend heavily on spatial representations to make navigation decisions [7]. Maps serve multiple purposes in autonomous systems:

- They provide spatial context for decision-making.
- They support route planning and obstacle avoidance.
- They act as a redundant sensing mechanism when sensors fail [7].

However, mapping in real environments remains challenging. Autonomous systems require highly detailed maps, and generating and updating them in real time is computationally expensive and technically difficult [7]. Additionally, real-world environments contain dynamic obstacles, occlusions, and sensor noise, all of which degrade localization accuracy. Because of these issues, perception systems typically rely on multiple sensors. Cameras provide rich semantic information but poor depth accuracy and are sensitive to lighting conditions, whereas LiDAR sensors provide accurate geometry but limited appearance information [6]. Combining sensing modalities improves robustness, which is why many autonomous robots integrate LiDAR, cameras, radar, and odometry.

Bird's-Eye-View and Environmental Representation

A widely adopted approach in modern autonomous navigation is to convert sensor data into a bird's-eye view (BEV) representation. BEV offers a unified spatial perspective of the environment and simplifies downstream navigation tasks such as path planning and control [5]. By projecting sensor information into a top-down map, the system can reason about obstacles and free space more effectively than using raw sensor data. BEV representations also form the basis of occupancy grids, which are commonly used in robotics to determine navigable areas. These representations are particularly suitable for maze-like or indoor environments because they allow the robot to reason about walls, corridors, and reachable paths.

Perception and Environmental Understanding

Autonomous navigation further depends on accurate environmental perception. Scene understanding such as identifying obstacles, pathways, and targets requires pixel-level interpretation of sensor data. Large, annotated datasets have been developed to train perception algorithms capable of recognizing objects,

estimating depth, and understanding spatial layouts [2]. Such perception is essential in robotics because navigation decisions depend on recognizing traversable areas rather than merely detecting obstacles. Despite recent advances, robust autonomy remains difficult in real environments due to sensor noise, lighting variation, occlusions, and unpredictable human behaviour [1]. Indoor environments further complicate perception due to multipath reflections and signal interference, which affect both vision sensors and wireless localization systems.

Relevance to Indoor Search and Rescue

Search-and-rescue environments present a unique combination of challenges:

- Lack of GPS signals
- Structural complexity
- Obstacle clutter
- Sensor interference
- Limited communication reliability

In such settings, robots must rely entirely on onboard sensing and local estimation rather than external positioning systems. SLAM provides environmental awareness, but it is fundamentally exploration-based and therefore time-consuming. Consequently, combining mapping with an external target-localization cue can significantly reduce search time. The use of signal-based localization (e.g., wireless ranging) offers a complementary approach to mapping. While SLAM identifies free space, wireless ranging provides directional information toward the target, allowing the robot to switch from exploration to goal-directed navigation. This hybrid strategy directly addresses a known limitation of autonomous systems. Although mapping allows safe navigation, it does not inherently provide knowledge of where a missing person is located.

Significance of the Proposed Work

Current research in autonomous driving and robotics has achieved significant progress in perception and mapping, yet fully reliable autonomous navigation in unstructured environments remains unsolved [1]. Most systems either explore the environment exhaustively or rely on prior maps, both of which are unsuitable for dynamic disaster scenarios.

The proposed project addresses this gap by integrating:

- SLAM-based mapping for environmental awareness
- Wireless ranging for target homing
- Probabilistic filtering for uncertainty handling

By combining mapping and signal-based localization, the system aims to reduce search time while maintaining navigation safety. This approach is timely because emergency response robotics is increasingly needed for disaster mitigation, infrastructure inspection, and hazardous environment operation. If successful, the work can contribute experimental data and practical insight into how low-cost sensing technologies can support indoor rescue missions.

Section 2 S/T methodology and associated work plan

2.1 Methodology

The proposed Search and Rescue Vehicle (SRV) will be developed following a modular simulation-driven methodology. The system will be divided into 4 parts: perception, localization and mapping, decision and planning, and actuation. Each part may be developed and validated independently of other parts, thus allowing for a modular design. Early validation using simulations and prototypes is prioritized to minimize potential risks and to ensure feasibility within given time frame.

2.1.1 Simulation Driven Design

Initially, a vehicle model will be implemented in ROS2 and simulated with Gazebo. This stage will focus on kinematic modelling, motor command validation, closed-loop velocity control, basic obstacle avoidance logic, and sensor data streaming. The goal of this simulation will be to validate actuation commands, verify coordinate framework transformations, and ensure proper integration of odometry and simulated sensor data before developing the system to an autonomous one. This simulation will help us understand the actuating capabilities of our vehicle and accordingly will aid us in coding the dynamic movement of the vehicle in response to sensory data.

We will then design constrained navigation routes in CARLA that represent the intended operating conditions: corridor-like segments, intersections, turns, cluttered passages, and dead ends. Each environment will be executed across multiple randomized trials (initial pose, illumination variation, etc.) to avoid tuning the pipeline to a single route.

CARLA will provide synchronized simulation outputs including:

- RGB camera frames (used for target identification training and evaluation)
- IMU measurements (angular velocity and linear acceleration)
- Wheel odometry / velocity estimates derived from simulated vehicle motion
- Event outcomes such as collisions and route completion

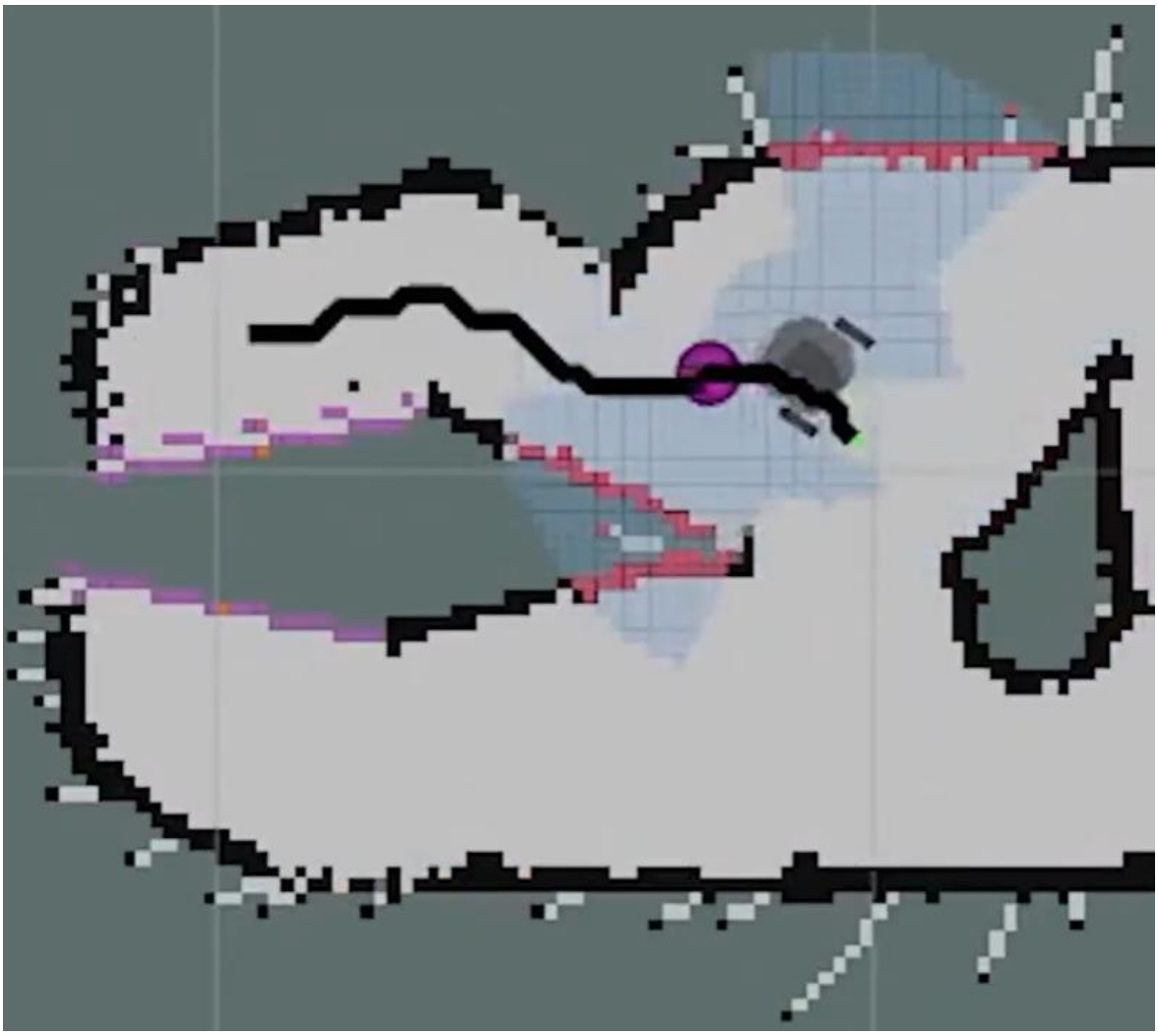


Figure 2: Sample simulation depicting object avoidance and real-time path planning

To obtain stable state estimates from noisy sensors, we will fuse IMU and wheel-odometry (or any other given sensor to be used in our project) signals using a probabilistic estimator (e.g., EKF/UKF). In simulation, filter parameters can be tuned and validated systematically because CARLA provides ground-truth for benchmarking. The estimator will track a state that includes heading/orientation, linear velocity, and position (as available from fused measurements), with optional bias terms if required for stability. We will tune process and measurement noise parameters under nominal and degraded sensing settings, including injected bias drift and sensor dropouts.

In addition to classical autonomy evaluation, Phoenix will implement a learning-based autonomy baseline based on a Bird's-Eye-View (BEV) representation derived from a single front-facing RGB camera. The goal is to predict short-horizon motion targets that allow the vehicle to make progress along a route while avoiding collisions. The BEV pipeline will follow a SimpleBEV-style structure: a lightweight image encoder backbone (chosen for feasibility), an image-to-BEV lifting/projection module that produces a BEV latent feature map, a prediction head that outputs a short-horizon plan, such as future waypoints in the self-frame (or an equivalent local motion target representation). Our training data will be generated in CARLA using an expert policy (CARLA autopilot or a scripted controller). Each recorded timestep will include camera frames, self-state (speed/heading), expert supervision (future waypoints and/or control targets), outcome labels (collision/route completion). The model will be trained to minimize waypoint/control prediction error. Evaluation will emphasize safety metrics (collision avoidance) and task completion rather than purely minimizing loss.

Once basic mobility and sensor fusion is validated, the simulation environment will be extended to include simulated (2D) LiDAR scans, wheel odometry, IMU data (if needed), and occupancy grid generation. We will utilize SLAM algorithms (e.g. Cartographer, slam_toolbox (ROS2)) to localize our system and map the surrounding environment. We will test our code in various virtual maze environments to evaluate map consistency, loop closure performance, pose estimation drift, and map resolution trade-offs. Future extensions may integrate learning-based visual odometry or graph-optimization backends such as g2o or ORB-SLAM3. The key deliverable being a generated occupancy grid, our system will utilize it to draw optimum paths from the starting point to the target destination using the A* algorithm and implement local obstacle avoidance with cost maps. Upon completion, the aforementioned algorithms and models will be implemented on our hardware system, where they will be further tuned to achieve a successful mission.

The simulation-driven methodology provides several key and novel advantages. Algorithmic instability can be identified before hardware deployment which reduces risk. The modular development of the vehicle allows the mapping, ranging, planning, and perception modules to be development in a parallel manner. The controlled environment makes it easy to pinpoint bottlenecks and failure points. All simulation experiments can be logged and repeated for quantitative comparison.

2.1.2 2-Stage Hardware Prototypes

The LiDAR needs to be imported and will take time to arrive, hence we decided to split the prototyping stage into 2.

Prototype 1: Ultrasonic Sensor Based Navigation and FTM Testing

The first prototype will be designed for the purpose of validating Wi-Fi FTM based on the IEEE 802.11mc protocol. It will feature reactive and rudimentary obstacle avoidance controlled via an onboard ESP32. Another ESP32 will be mounted atop the SRV, and its mission will be to communicate with the beacon ESP planted in the maze, and guide the prototype. The goal of this prototype is to implement the basic control logic for the motors, and to log the measurements of distance between the beacon and the SRV based on Wi-Fi FTM.

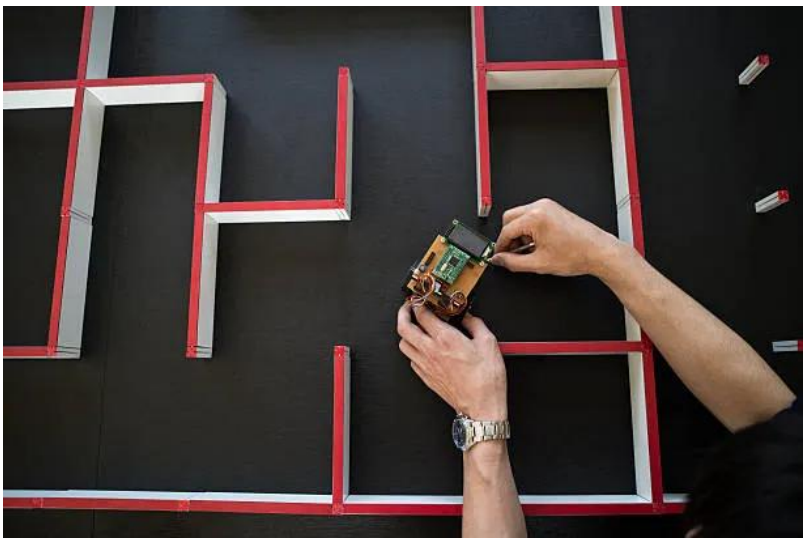


Figure 3: Stock image showing a small Arduino car; Prototype 1 may resemble it. [11]

Prototype 2: LiDAR Based SLAM

Upon arrival of the LiDAR, a second prototype will be made. This prototype will now utilize the wheel encoders in the motors and the LiDAR, possibly an IMU as well, to perform mapping and localization of the vehicle. SLAM will enable us to generate an occupancy grid of the maze, estimate our position by scan/motion matching and odometry fusion using one of the industry standards, e.g.: Extended Kalman Filter. It will also allow us to detect obstacles, construct a motion path, and enable the deployment of our fully autonomous navigation stack.

2.1.3 Wi-Fi FTM Based Localization

One of the key objectives of this project is the evaluation of the viability of Wi-Fi Fine Timing Measurement for target localization within a maze as well as collecting statistics based on it. The set up will be simple, consisting of a single ESP32 S3 on the SRV and another S3 acting as a beacon in the maze. The ESP on the SRV will attempt to ping the second ESP and request FTMs.

The main feature to be used for target localization is Round Trip Time (RTT). RTT is the measurement, in time, between the sending and receiving of packages from one Wi-Fi device to another [10]. It functions as follows, one device sends a data packet containing the sending time to a second device, the second device logs the arrival time of the message, then sends a response packet containing its own departure time as well as the arrival time of the first packet. The first device can then compute the round-trip propagation delay and estimate the distance between devices using the speed of light.

Upon receiving the dataset from prototype I, statistical analysis to determine how to best filter the data will be done. This will be followed with outlier rejection algorithm development to mitigate multipath effects. In prototype I, once the above is done, the on-board ESP will send a signal to the SRV to move. However, in prototype II onwards, the ESP will only act as a relay, sending the RTT/measurements to the raspberry pi where a full particle filter will constantly be running.

A particle filter is a probabilistic state estimation algorithm that represents all potential positions of a target as “particles” [12]. It starts with many particles distributed across space. It then takes in the distance measurements, sees which particles are at that distance away, then updates their likelihood using some form of probabilistic model; The model must use stats collected from the dataset of prototype I. After a while, most particles have a probability of almost 0 and get removed, leaving only the approximate true location. By the end of this, we will be able to collect data on FTM such as localization error versus true distance, effect of range on accuracy, impact of surface types and quantity, and more.

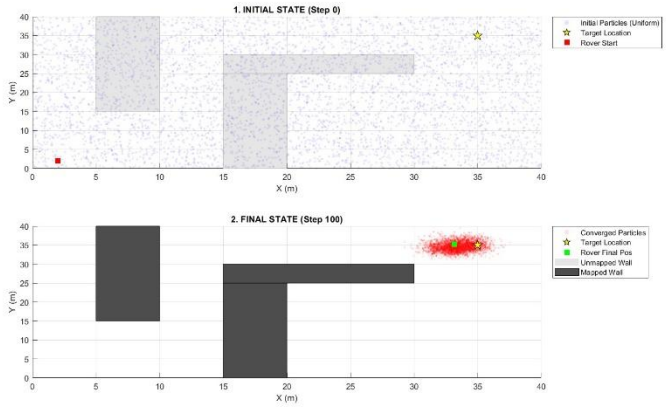


Figure 4: Particle Filter Simulation Showcasing SRV motion

2.1.4 Target Verification

2.1.4.1 Objective

After Phoenix navigates to a candidate target region (estimated via SLAM and/or RTT homing), it performs a target verification step to confirm the presence of the intended target before declaring mission success using its ESP32 OV2640 (2MP) Camera Module. Our primary target is a human-shaped object; accordingly, the verification model will be trained/fine-tuned for reliable human-figure detection under our indoor test conditions. At the same time, verification is implemented as a general confirmation stage. Phoenix's success criterion is to reach the candidate location and confirm the intended target, even if the exact target appearance/signature is represented differently during controlled experiments.

2.1.4.2 Hardware and data flow

A ROS 2 perception node processes frames locally and publishes detections (bounding boxes + confidence) along with a target_confirmed signal. Verification is triggered only within an inspection radius of the candidate location (e.g., 0.5–1 m) to keep compute usage manageable during exploration.

2.1.4.3 Verification Method

The perception node runs a lightweight target detector and applies multi-frame confirmation to avoid single-frame false positives. Confirmation requires (i) confidence above a tuned threshold and (ii) persistence across multiple frames using K-of-N voting (or other methods), optionally with bounding-box stability checks. During verification, Phoenix slows or stops (and may briefly scan/rotate) to improve reliability.

2.1.4.4 Training process (transfer learning)

We will fine-tune a pretrained lightweight detector using images/video collected with the camera in representative indoor environments. Data will include positive samples (person/human-shaped figure at varied distances/poses/occlusions) and negative samples (furniture silhouettes, posters, reflections) to reduce false confirmations. After annotation and dataset splitting, we will fine-tune with early stopping, evaluate using precision/recall and verification success rate, then export to an efficient inference format (e.g., ONNX/TFLite) with runtime tuning (resolution/FPS/thresholds) for embedded performance.

2.1.4.5 Integration and Behavior

If verification succeeds, Phoenix stops and reports the location as "target found." If verification fails within a short window, Phoenix resumes exploration and continues collecting measurements.

2.2 Work Package Descriptions

Work package number	**1**	Start date or starting event:		**Week 1-3**	
Work package title	**Background Research, BoM**				
Participant number	1	2	3	4	5
Participant name	Ömer Maraş	Mohamad Alomar	Abdul Rahman Abu Audeh		
Weeks per participant	**3**	3	3		

Objectives

- Conduct literature review on Autonomous driving, Wi-Fi FTM, SLAM, and particle filtering
- Study IEEE 802.11mc RTT framework
- Research possible simulation setups
- Define system architecture and BoM

Description of work

- **T1.1** (w1-w2) Literature review

Study the recent findings and research in the domains of autonomous driving, Wi-Fi FTM, SLAM, and particle filtering.

- **T1.2** (w1-w3) Homing Feasibility

Study the possible signals to be used for homing including UWB, LBE, LORA, etc.

- **T1.3** (w2-w3) Research possible simulation setups

Investigate viable simulation environments for SLAM, autonomous path planning, obstacle avoidance, etc.

- **T1.4** (w2-w3) Define system architecture and BoM

Create a system overview and create the bill of materials.

Deliverables

D1.1 (w1) Literature Review Summary
D1.2 (w2) Project Registration Form
D1.3 (w3) System Diagram
D1.4 (w3) Project Proposal

Milestones

M1.1: Finalized System Design
M1.2: Clear Simulation Roadmap

Work package number	**2**	Start date or starting event:		**Week 4-5**	
Work package title	**Complete Prototype I**				
Participant number	1	2	3	4	5
Participant name	Ömer Maraş	Mohamad Alomar	Abdul Rahman Abu Audeh		
Weeks per participant	**2**	2	2		

Objectives

- Simulate autonomous obstacle avoidance
- Build Prototype I
- Implement RTT logging framework

Description of work

- **T2.1** (w4) Simulate autonomous obstacle avoidance

Test basic vehicle manoeuvrability and control in GAZEBO and ROS2 simulation environments. Additionally simulate base autonomous path planning, and obstacle avoidance in CARLA.

- **T2.2** (w4-w5) Build Prototype I

Acquire hardware components and implement motor control and basic obstacle avoidance using ultrasonic sensors.

- **T2.3** (w5) Implement RTT logging framework

Setup platform.io, model the control systems, and flash the collection firmware onto the ESP.

Deliverables

D2.1 (w4) Running Simulation
D2.2 (w5) Functional Prototype I
D2.3 (w5) Logged RTT Dataset

Milestones

M2.1: Initial Simulation
M2.2: Prototype I Operational
M2.3: Initial RTT Dataset

Work package number	**3**	Start date or starting event:		**Week 6-7**	
Work package title	**RTT Data analysis and SLAM Simulations**				
Participant number	1	2	3	4	5
Participant name	Ömer Maraş	Mohamad Alomar	Abdul Rahman Abu Audeh		
Weeks per participant	**2**	2	2		

Objectives

- Analyse RTT oscillation behaviour
- Develop preprocessing and outlier rejection methods
- Model measurement error characteristics
- Determine viability of FTM for homing
- Code SLAM simulation

Description of work

- **T3.1** (w6) RTT Dataset Cleaning and Statistical Analysis

Process the collected RTT dataset to remove corrupted measurements and identify outliers. Perform statistical analysis of oscillation behaviour, including variance, bias, drift. Visualize temporal stability and distance-dependent error trends.

- **T3.2** (w6-w7) Measurement Model Development

Develop a probabilistic measurement model for Wi-Fi FTM ranging based on empirical data. Estimate noise distribution parameters and construct a likelihood function suitable for particle filtering or Bayesian localization frameworks.

- **T3.3** (w6-w7) FTM-Based Localization Feasibility Study

Using the derived measurement model evaluate convergence behaviour, positional error, and robustness to determine viability for homing applications.

- **T3.4** (w6-w7) SLAM Simulation and Integration

Develop and test SLAM simulation using LiDAR-based mapping in the ROS2/Gazebo environment. Integrate localization and mapping modules and evaluate map consistency, loop closure behaviour, and localization drift under controlled obstacle scenarios.

Deliverables

D3.1 (w6) Cleaned and Annotated RTT Dataset

D3.2 (w6) Statistical Error Analysis Report
D3.3 (w7) Validated FTM Measurement Model
D3.4 (w7) SLAM Simulation Results and Performance Evaluation Report

Milestones
M3.1: Approved FTM viability decision
M3.2: Measurement model finalized
M3.3: Tested localization and mapping for autonomous mission

Work package number	**4**	Start date or starting event:		**Week 8-10**	
Work package title	**Complete Prototype II**				
Participant number	1	2	3	4	5
Participant name	Ömer Maraş	Mohamad Alomar	Abdul Rahman Abu Audeh		
Weeks per participant	**3**	3	3		

Objectives
- Integrate LiDAR, wheel encoders, and Raspberry Pi
- Implement SLAM (ROS2-based) on Prototype II
- Implement particle filter localization
- Integrate full navigation stack

Description of work
- **T4.1** (w8) Hardware Integration and System Assembly

Integrate LiDAR sensor, wheel encoders, and Raspberry Pi onboard computing unit onto Prototype II. Establish power distribution architecture, and sensor mounting calibration. Validate encoder feedback consistency and LiDAR data streaming within the ROS2 framework.

- **T4.2** (w8-w9) SLAM Implementation and Real-Time Mapping

Deploy ROS2-based SLAM (e.g., Cartographer or equivalent) on the Raspberry Pi. Configure sensor fusion between LiDAR and wheel odometry. Tune parameters for scan matching, map resolution, and loop closure detection. Validate real-time map generation and evaluate mapping accuracy against controlled indoor layouts.

- **T4.3** (w9) Particle Filter Localization Integration

Implement a particle filter–based localization module. Evaluate localization convergence, stability, and robustness under sensor noise.

- **T4.4** (w9-w10) Full Navigation Stack Deployment

Integrate global and local planners within the ROS2 navigation stack. Configure costmaps, obstacle inflation layers, and recovery behaviours. Validate autonomous exploration, obstacle avoidance, and goal-directed navigation within indoor test environments.

Deliverables
D4.1 Fully integrated SLAM-capable SRV
D4.2 Real-time particle filter implementation
D4.3 Integrated navigation system

Milestones
M4.1: Successful map generation
M4.2: Autonomous exploration achieved
M4.3: Target homing capability functional

Work package number	**5**	Start date or starting event:	**Week 11-14**

Work package title	**Target Recognition Integration and System Finalization**				
Participant number	1	2	3	4	5
Participant name	Ömer Maraş	Mohamad Alomar	Abdul Rahman Abu Audeh		
Weeks per participant	**4**	4	4		

Objectives

- Implement vision-based target detection
- Integrate perception with navigation stack
- Validate autonomous mission workflow
- Perform system-level testing and performance evaluation
- Finalize documentation and demonstration

Description of work

- **T5.1 (**w11) Target Recognition Model Development and Training

Develop and train a computer vision model for target detection using a curated dataset. Perform data preprocessing, augmentation, and annotation. Optimize model size and inference speed for deployment on Raspberry Pi (or edge device). Evaluate precision, recall, F1-score, and inference latency.

- **T5.2** (w11-12) Real-Time Inference Integration

Integrate the trained model into the onboard system. Implement real-time image acquisition and inference pipeline. Establish communication between perception module and navigation stack to trigger behavioral changes upon target detection.

- **T5.3** (w12-13) System-Level Testing and Performance Evaluation

Conduct trials in the maze environments. Record metrics including target detection accuracy, false positive/negative rate, end-to-end mission success rate, localization error at detection, mission completion time. Analyse failure cases and refine parameters.

- **T5.4** (w12-14) Final Optimization and Documentation

Optimize system latency and stability. Finalize architecture diagrams, performance tables, and demonstration materials.

Deliverables

D5.1 (w11) Trained and validated target recognition model
D5.2 (w11) Integrated perception-navigation system
D5.3 (w12) Autonomous end-to-end SAR demonstration
D5.4 (w12) Final technical report and evaluation results

Milestones

M5.1: Real-time target detection achieved
M5.2: Fully autonomous search-detect-homing mission successful
M5.3: Final system validation complete

2.3 Demonstration

The final demonstration will be of a fully autonomous SAR task conducted across the three different mazes described prior: baseline, reflective, and Wi-Fi congested. The SRV will begin from a known start position and must localize, map, identify and reach a Wi-Fi beacon acting as a rescue target without external intervention. Qualitative and Quantitative measures will be used to analyse the system.

Localization Accuracy: The primary metric will be the measure of the Euclidean distance between the estimated target location and ground truth position at the start, middle, and end of the maze. That is, when the full distance is “d”, “d/2”, and 0. The mean error, standard deviation, and gravity of error will be reported for each maze across multiple trials.

SLAM-Only Baseline: To test the actual benefit of using RTT homing, performance metrics will be collected from multiple FTM-less runs. Some of these metrics will include time to reach target, number of nodes expanded, total path length, percentage of area explored. This will give a quantitative measure of the improvements FTM RTT provides.

SLAM Accuracy: Accuracy in the depiction of the generated environmental map using the LiDAR sensor relative to ground-truth measurements.

Trilateration vs Particle Filter: To test the benefit of a probabilistic based approach, particle filter, a simple geometric trilateration method (or LSE) will be used as a comparison. Some of the metrics collected will include localization error, convergence rate, and outlier rejection ability. We aim to prove the particle filter is more robust for such a scenario.

Cross-Maze Statistics: We aim to collect data on the degradation of performance across the 3 different mazes. Some of the metrics collected will include RTT variance, localization error, convergence time and rate, rescue time, oscillation of system, outlier rejection performance, and others to be able to showcase environmental effect on performance.

Target Confirmation: The accuracy of the object/target detection model in correctly identifying the target, as evaluated against the labelled training and validation datasets.

By the end, we will have several quantitative stats on usability of RTT for SAR performance showcasing its effectiveness (or lack thereof). We will also have an SRV capable of rescuing targets lost within a maze. The final demonstration on senior design day will include several videos of runs in the mazes, some data tables showing statistics and conclusions, as well as an actual prototype on the table. Potentially, if given enough space, we may build a mini-maze and allow people to place the beacon themselves, and the SRV will autonomously drive to it.

2.4 Impact

Phoenix aims to deliver a simulation-driven autonomous navigation stack for indoor, GPS-denied, search-and-rescue-inspired scenarios, and to quantify how much additional benefit (if any) Wi-Fi FTM/RTT homing provides in such environments. The main impact of this project will be the evaluation of the feasibility of using Wi-Fi FTM in indoor localization in structured maze-like environments. This project hopes to contribute controlled experimental data and statistical analysis on localization error, multipath distortion, and more statistics on RTT. If decent localization accuracy and rescue time can be achieved, this work may encourage further use of RTT enabled devices in emergency situations and encourage people to keep their devices on them during emergencies. We also hope it motivates further research into the use of RTT and possibly mesh network RTT for SAR operation. Lastly, a prototype for indoor SAR support will be developed and tested in structured indoor environments to guide future SAR robotics development.

2.5 Risk analysis

Risk 1 – RTT oscillation and FTM reliability

Indoor Wi-Fi FTM measurements can be unstable due to multipath, bias, and congestion, causing large distance oscillations and unreliable homing.

Mitigations:

- Preprocessing and measurement modelling: apply outlier rejection, variance gating, and environment-specific calibration/likelihood tuning.
- Robust probabilistic estimation: tune particle count/resampling and add recovery behaviour when estimates collapse to reflections; constrain estimates using SLAM occupancy information when

possible.

- Staged motion strategy: collect initial measurements during a short non-homing phase to stabilize early estimates before committing to homing.

If RTT is not reliable enough for closed-loop homing, we complete the project with the SLAM-only baseline as the primary navigation method and present FTM as an evaluated add-on rather than a dependency.

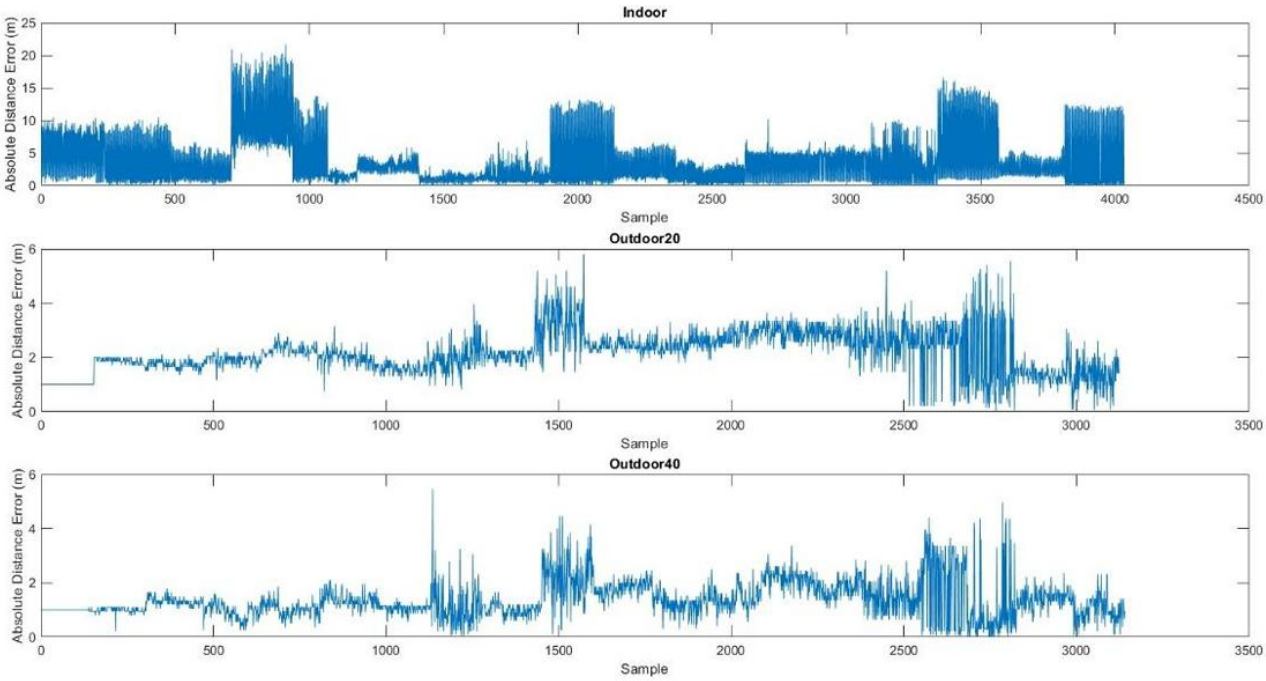


Figure 5: Graphs showcasing distance error between actual distance and measured distance using RTT under 3 separate conditions [13].

Risk 2 – Obstacle detection blind spots

A 2D LiDAR may miss ground-level hazards or small obstacles, increasing the chance of collisions or stuck states.

Mitigations:

- Add simple near-ground sensing (ultrasonic) and enforce conservative speed limits in cluttered areas.
- Improve sensor placement and validate detection coverage early in controlled tests.
- Add fail-safe behaviours (stop, replan, reverse, manual override) when localization confidence drops or the robot becomes stuck.

Risk 3 – System integration and timing/communications

Multi-component integration (microcontroller, onboard computer, ROS nodes) may introduce delays, dropped messages, or timing mismatches that affect closed-loop behaviour.

Mitigations:

- Define clean interfaces and message contracts early; log timestamps end-to-end; test modules independently before full integration.
- Keep the mission-critical loop on the main compute where possible and treat external modules as input providers when feasible.

Risk 4 – Hardware delays or failures

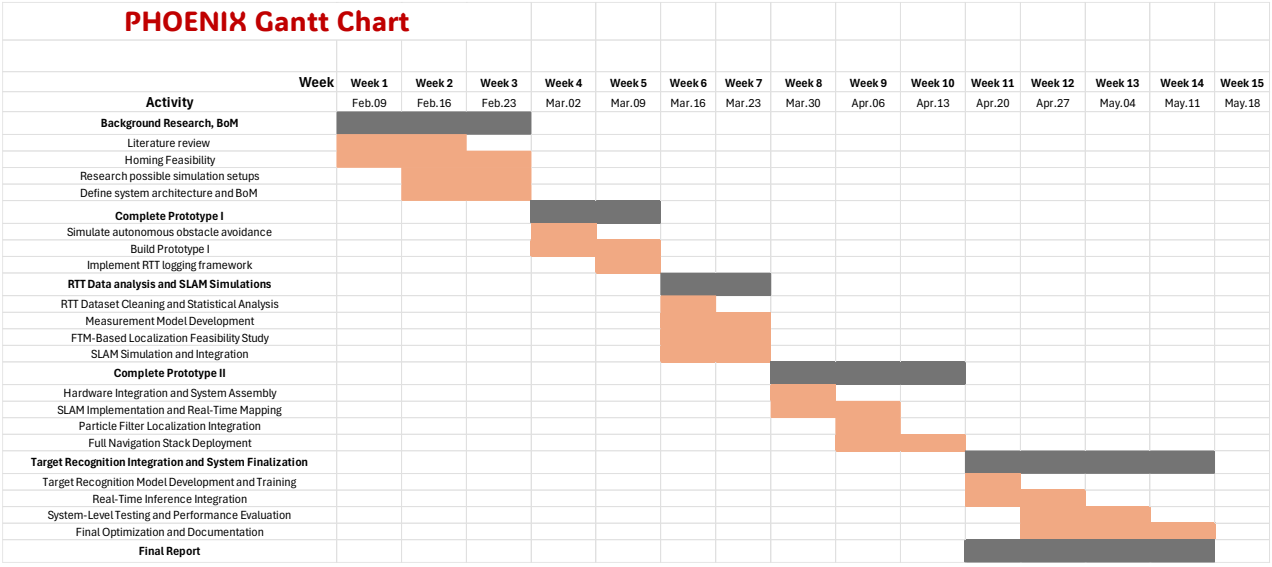
Shipping delays, sensor faults, or power issues can compress the hardware timeline.

Mitigations:

- Maintain a simulation demo path that satisfies the MVP even if hardware is delayed.

- Use staged prototypes to avoid blocking progress.

2.6 Gantt Chart



2.6.1 Task Distribution Matrix

PHOENIX Task Distribution Matrix

ÖM: Ömer Maraş, MA: Mohamad Alomar, AA: Abdul Rahman Abu Audeh

Week	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14
Activity	Feb.09	Feb.16	Feb.23	Mar.02	Mar.09	Mar.16	Mar.23	Mar.30	Apr.06	Apr.13	Apr.20	Apr.27	May.04	May.11
Background Research, BoM														
Literature review	ÖM, MA, AA	ÖM, MA, AA												
Homing Feasibility	AA	AA	AA											
Research possible simulation setups		ÖM, MA	ÖM, MA											
Define system architecture and BoM		ÖM, MA, AA	ÖM, MA, AA											
Complete Prototype I														
Simulate autonomous obstacle avoidance				ÖM, MA										
Build Prototype I				ÖM, MA, AA	ÖM, MA, AA									
Implement RTT logging framework					AA									
RTT Data analysis and SLAM Simulations														
RTT Dataset Cleaning and Statistical Analysis						AA								
Measurement Model Development						AA	AA							
FTM-Based Localization Feasibility Study						AA	AA							
SLAM Simulation and Integration						ÖM, MA	ÖM, MA							
Complete Prototype II														
Hardware Integration and System Assembly								ÖM, MA, AA						
SLAM Implementation and Real-Time Mapping								ÖM, MA	ÖM, MA					
Particle Filter Localization Integration									AA					
Full Navigation Stack Deployment									ÖM, MA, AA	ÖM, MA, AA				
Target Recognition Integration and System Finalization														
Target Recognition Model Development and Training											ÖM, MA			
Real-Time Inference Integration											ÖM, AA	MA, AA		
System-Level Testing and Performance Evaluation												ÖM, MA, AA	ÖM, MA, AA	
Final Optimization and Documentation												ÖM, MA, AA	ÖM, MA, AA	ÖM, MA, AA
Final Report														

Section 3 Economical and Ethical Issues

3.1 Ethical and Safety Considerations

This project involves a mobile robot platform and sensing; therefore, safety is a primary ethical requirement. We will follow safe electrical practices (battery handling, insulation, appropriate motor driver usage) and safe mechanical practices (secured mounting, protected rotating parts). We will also implement operational safety measures such as conservative speeds during indoor tests, an emergency stop/kill procedure, and testing in controlled environments away from bystanders.

3.2 Legality and Responsible Experimentation

All wireless experiments will comply with IEEE standards [14] and local telecommunications regulations; no illegal interference or jamming will be used. For congested scenarios, we will only use legal, non-disruptive methods (e.g., permitted access points and natural Wi-Fi traffic) and avoid practices that could interfere with third-party networks.

3.3 Privacy and Data Handling

RTT-based ranging does not require collecting personal user data beyond the minimal exchange needed for measurement. Any logged data will be limited to what is required for evaluation (timestamps, RTT samples, robot state) and stored securely. If a camera is used for target verification, tests will use non-human targets (e.g., markers/objects), avoid capturing identifiable imagery of people, and store data locally for the minimum duration required for analysis.

3.4 Dual-use Considerations

Autonomous navigation technology can be repurposed outside SAR contexts. To address dual-use concerns, we scope the system explicitly to controlled SAR-support demonstrations (maze navigation and safe indoor testing) and do not develop capabilities intended for surveillance or harmful applications.

3.5 Economical Constraints and Feasibility

The project is designed to be economically feasible by prioritizing software-first development using simulation to reduce reliance on costly hardware iterations; using commodity components and limiting custom fabrication to what is necessary for mounting and stability; and structuring features as MVP plus stretch goals so that time and budget constraints do not jeopardize core deliverables (robust navigation stack plus evaluation).

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